UNSW MARKETING ANALYTICS HACKATHON 2026

Beyond a good story

What 1.45 million Kiva loan descriptions
can and cannot tell us about funding speed







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Photos: Kiva; Muhammad-Taha Ibrahim and David Geneugelijk on Unsplash

Text carries forecast information, but no writing tip is proven

Summary across the explanatory models, robustness checks and forecast benchmark

EXPLANATORY MODELS

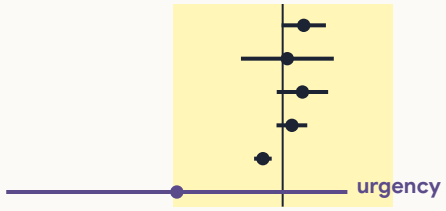
64 of 120

Uncertainty first

Yellow squares: significant terms that lost significance once errors were clustered by 48 countries.

EQUIVALENCE TEST, EXPLORATORY

±5%

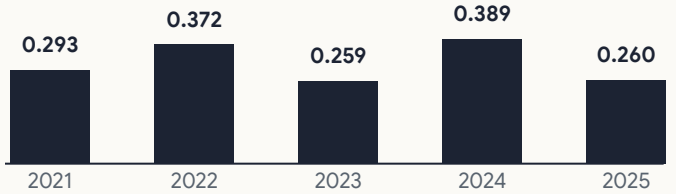


Most framing links are small

Five framings stay inside the ±5% band per extra mention per 100 words; urgency does not.

FORECAST BENCHMARK

5 of 5 years



Text adds forecast information

Text lowered linear forecast error every year, by 0.26–0.39 days; the median is about 7 hours.

Next decision: validate the text measures, then test one truthful prompt with partners.

Validate measures | Close data gaps | Pilot one prompt

Agenda

01

Problem

What are we measuring, and why does it matter?

- Editable fields 4
- The outcome 5
- A slower market 6

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Approach

How did we test the claims?

- Three evidence layers 7

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Evidence

What holds up, and how well?

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Next steps

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The description is one of the few listing fields still editable

Context: what is fixed and what is editable when a Kiva loan is listed

Fixed by the time a loan is listed	Still editable
Loan amount and term	**Description text** 2
Sector and purpose	Loan use statement
Country and partner	Photo
Borrower gender and group size	

1 Why the question matters

Partners and Kiva staff coach descriptions; if framing mattered, coaching would be a cheap lever.

2 What we asked

Which narrative choices are associated with faster funding, for whom and when?

The outcome is platform funding time, not borrower waiting time

Descriptive audit of all records

1,453,840 ②

valid completed durations, 2016–2025

96.4%

disbursed to the borrower before listing (median 24.2 days earlier)

Typical sequence for a loan disbursed before listing



Scope: 1,453,846 input records; 6 negative durations excluded; 1,637 refunded records included.

① Who waits

For most loans the partner has already paid the borrower. Platform time plausibly reflects replenishing partner capital; it is not measured directly.

② Who is missing

Only loans that eventually funded are in the data, so we cannot study whether a loan funds at all.

Same-day funding fell from 46.0% to 30.3% and has not recovered

Share of loans fully funded within 24 hours by period; descriptive

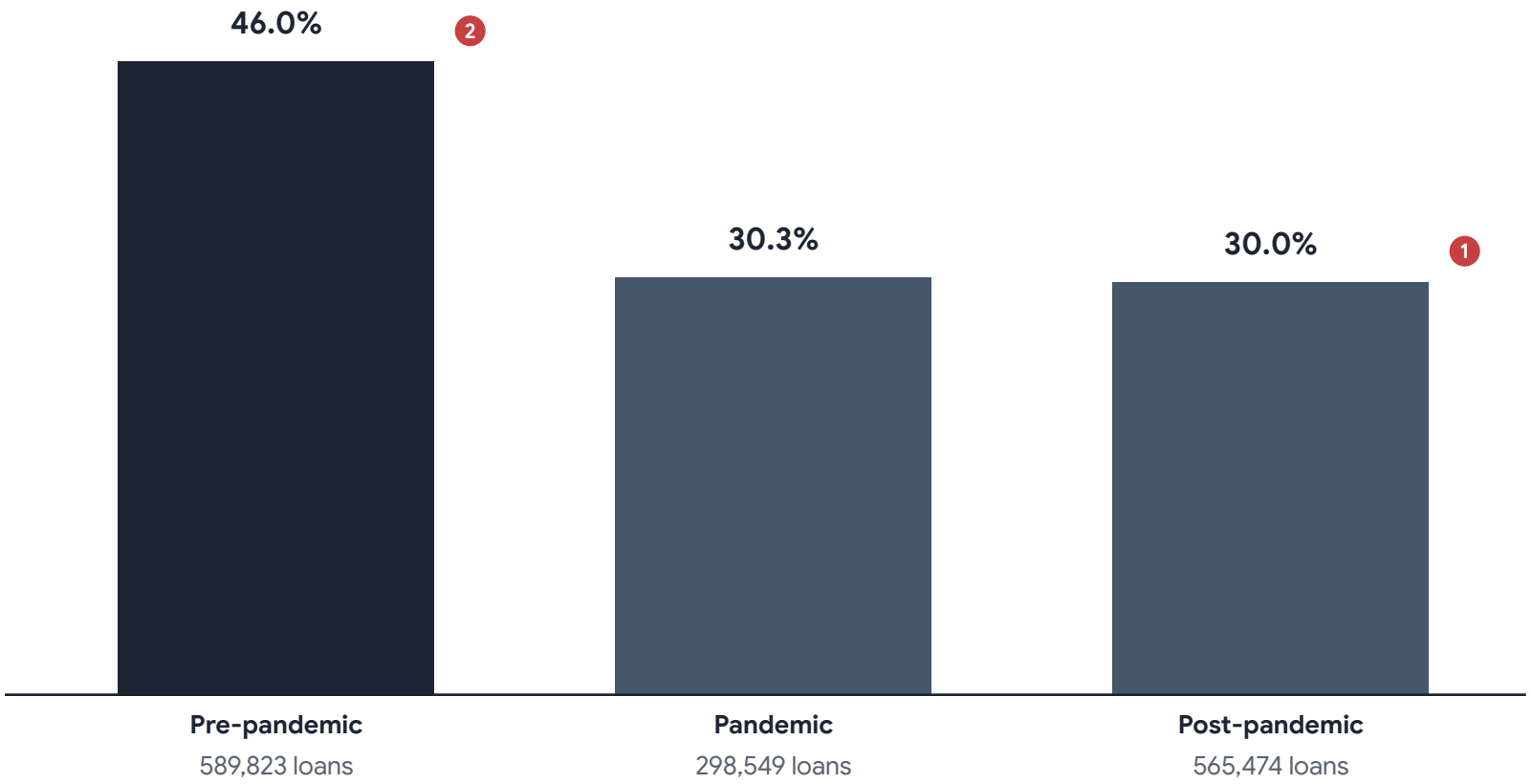


Figure 1. Share of completed loans funded within 24 hours. Bars start at zero.

1 A slower market

The drop persisted for four years after the disruption.

2 Why it matters here

Period is a control in every model; narrative effects are judged within periods.

Three layers of evidence, each answering a different question

Explanatory models, robustness checks and a forecast benchmark

Explanatory models

Question

Which features are associated with speed, and how sure are we?

Method

OLS and GLM; HC3 versus country-clustered errors; 2024–2025 chronological holdout

Robustness checks

Question

Do associations survive richer controls?

Method

Country-clustered OLS and LPM with and without a programme-text proxy; BH adjustment

Forecast benchmark

Question

How much does text add to a structure-only forecast?

Method

Matched feature blocks on identical rows; as-of training; five annual windows 2021–2025

Equivalence test: are the robustness-check associations small enough to bound?

Large row counts did not remove country-level uncertainty

Explanatory models: HC3 versus 48-country clustered errors

64 of 120

significant terms lost significance under country clustering

p = 0.49 2

urgency, duration model, clustered (was p < 0.001 under HC3)

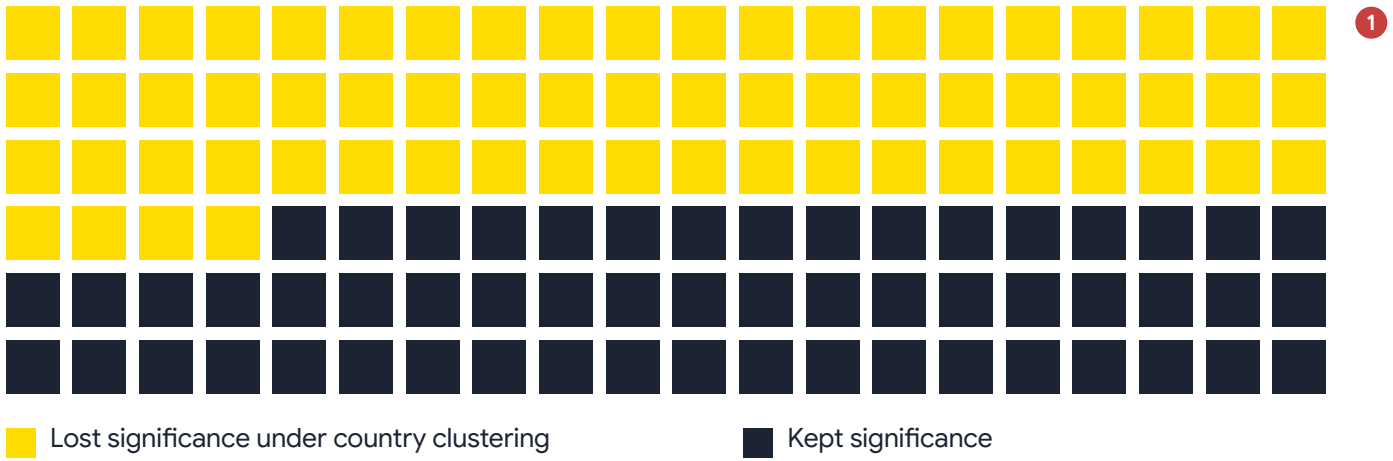


Figure 2. Each square is one term significant under HC3, across both explanatory models.

1 Why it happens

Loans in one country share context, so 1.45 million rows behave like far fewer independent observations.

2 What it does not mean

Losing significance is not evidence of no effect; the next slide tests that directly.

Most framing associations are small; urgency remains unresolved

Equivalence test of the robustness-check coefficients; exploratory, bound set after the estimates

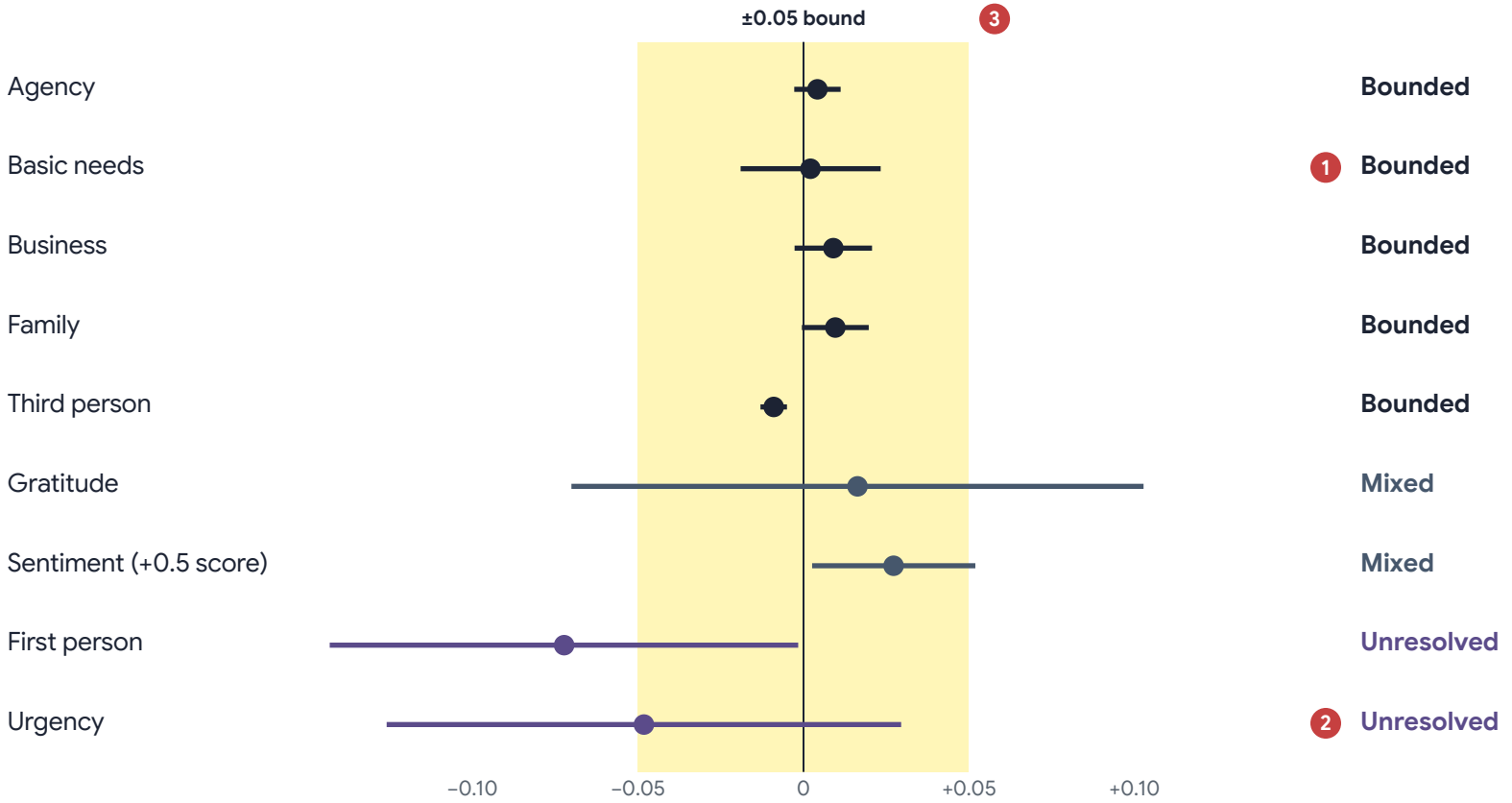


Figure 3. Change in log(1 + days) per extra mention per 100 words (sentiment per +0.5 score), 90% interval, expanded specification. Status covers both specifications.

1 Small, not just non-significant

Five framings stay inside ±5% of funding time in both specifications.

2 Too noisy to judge

Urgency and first-person framing could still be linked to 6–14% faster or slower funding per mention. Each is absent from over 98% of records.

3 Caveat

The bound was chosen after seeing the estimates; a confirmatory test needs one agreed in advance.

Regional family-framing patterns rest on two countries each

Explanatory models: within-region slopes with country-clustered and heuristic few-cluster intervals

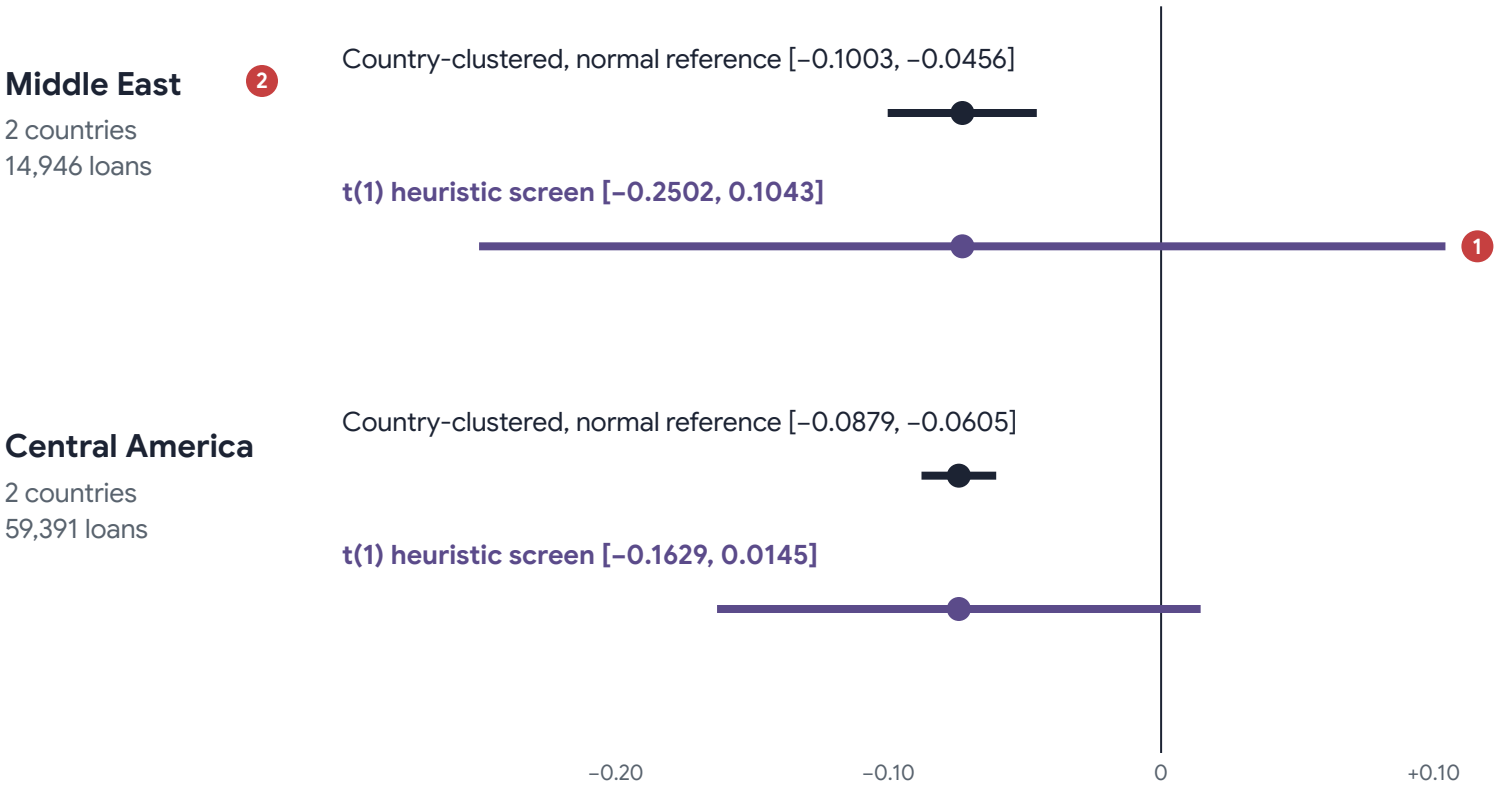


Figure 4. Change in log(1 + days) per extra family mention per 100 words. Dots are the slope estimates.

1 Two countries is not a pattern

With two clusters, between-country uncertainty cannot be estimated well; the heuristic intervals cross zero.

2 No regional advice

These are descriptive patterns, not grounds for country or regional writing rules.

Few narrative associations remain after programme controls

Robustness checks: country-clustered, BH-adjusted within specification

Term and outcome	Expanded	Programme-adjusted
Third person, log(1 + days)	−0.0090, BH p 0.012	−0.0081, BH p 0.061
Business, 24-hour (pp)	not significant after BH	−0.258 pp, BH p 0.006

0.012 → 0.061 [1]

third-person BH p, expanded to programme-adjusted

−0.258 pp [2]

business framing, same-day rate, programme-adjusted

Note: Per extra mention per 100 words. BH-adjusted within each specification.

1 Small and mixed

Third-person framing keeps its size but rises to BH p 0.061; business framing is the one term below BH 0.05.

2 Still associations

Programme text is an unvalidated proxy; neither result shows what editing would do.

Adding text lowered forecast error in 5 of 5 years

Forecast benchmark: linear model, structure only versus structure plus text, identical evaluation rows

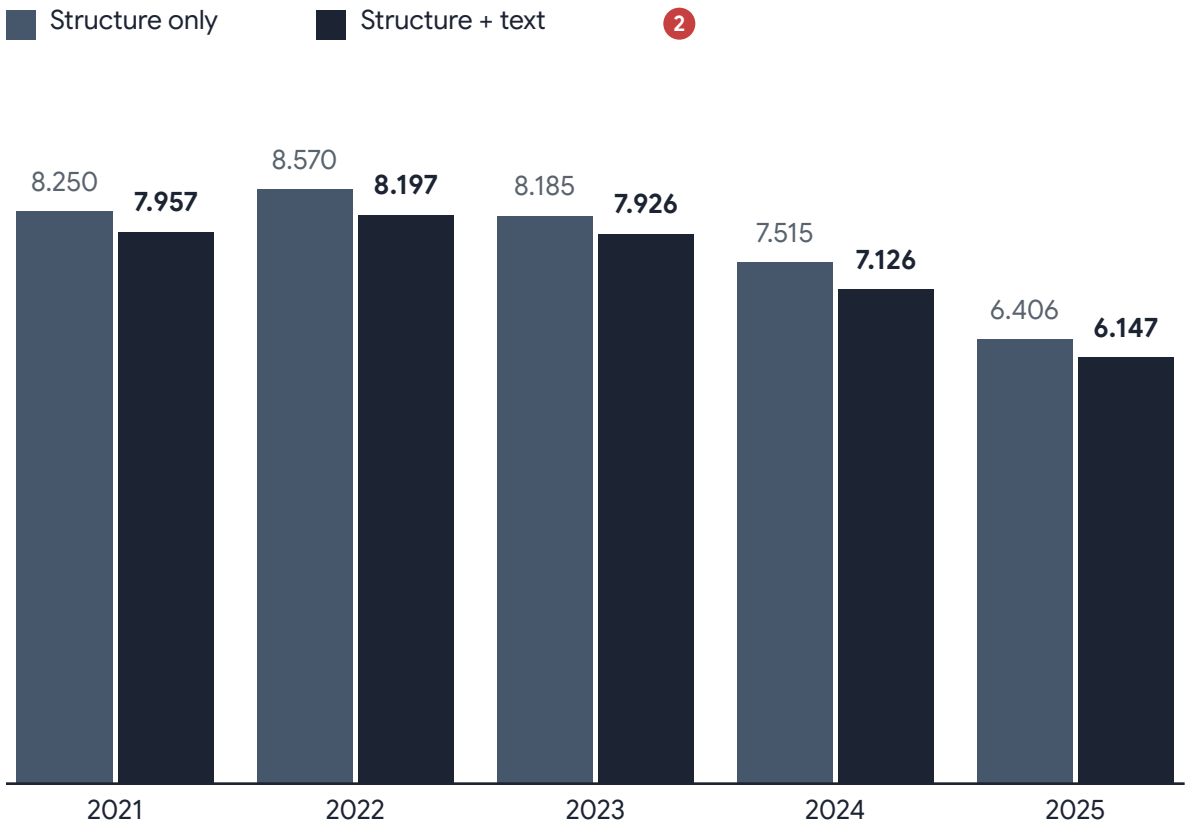


Figure 5. MAE in days, lower is better. Linear model on identical evaluation rows. Bars start at zero.

0.29 days 1

median annual error reduction

5 of 5

years with lower error

728,640

evaluation records

1 Text is informative

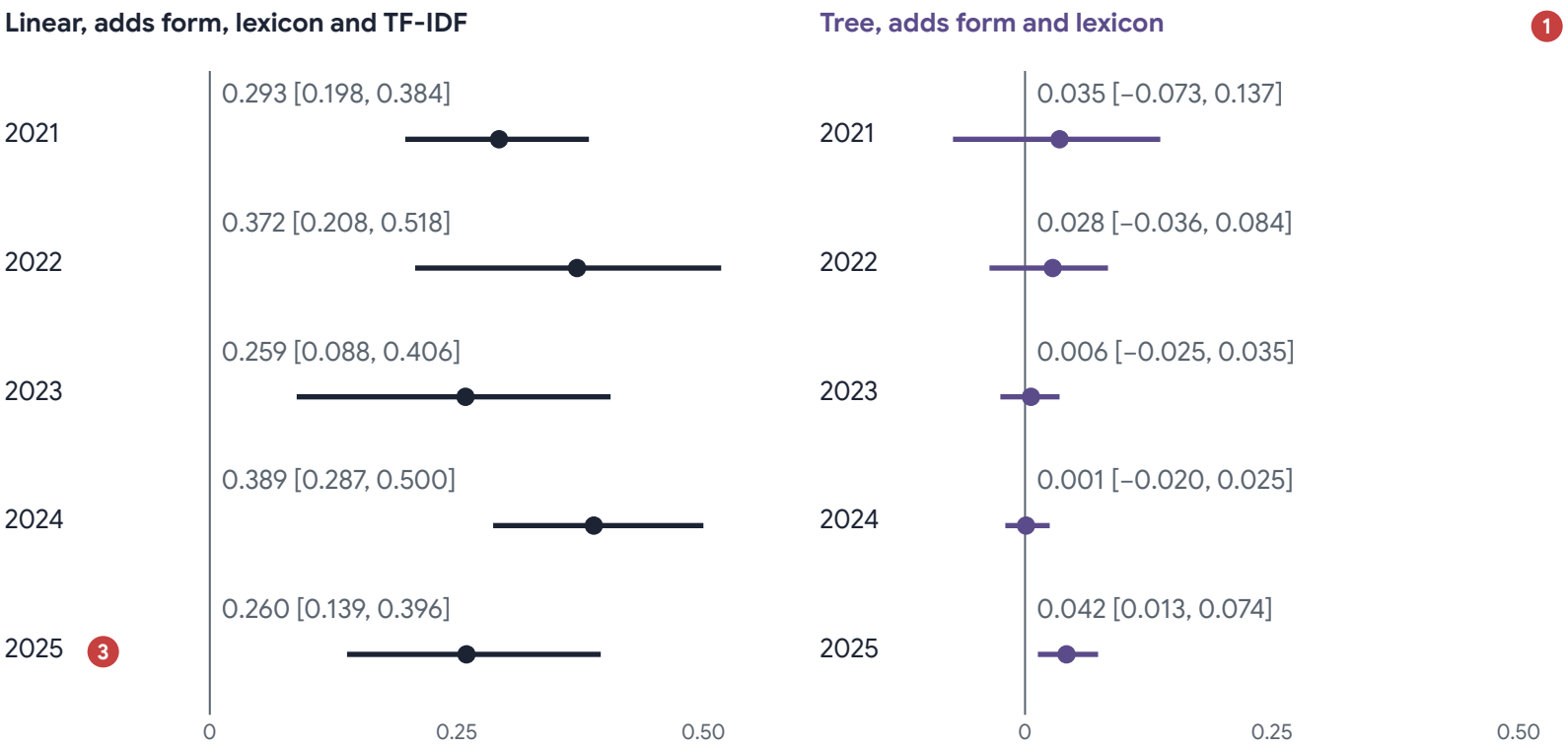
Gains of 0.26–0.39 days on 728,640 evaluation records.

2 Not an editing effect

Text may encode purpose, place or programme templates, not only persuasion.

The gain is modest and depends on the model

Forecast benchmark: annual MAE gain with month-resampled 95% intervals



2 Country resampling, not shown: intervals include zero in 4 of 5 linear and 3 of 5 tree years.

Figure 6. MAE gain in days with month-resampled 95% intervals; positive is better. Both panels share one scale.

1 Model matters

The tree model, without TF-IDF, gains at most 0.04 days.

2 Countries matter

Country resampling includes zero in 4 of 5 linear years. Capping extreme forecasts, an exploratory check, leaves zero in 1 of 5.

3 Metrics disagree

In 2025 text improved error and calibration but AUC fell from 0.893 to 0.888.

What the evidence supports, and what it does not

Synthesis across all layers

Supported	Not established
Text adds forecast information in a linear model	That editing a description speeds funding
Most framing associations are small	Any country or regional writing rule
Platform funding time is not borrower waiting time for most loans	Effects of urgency or first-person framing
	Performance on loans that never funded

Three gates before any writing advice

Proposed next research for Kiva and participating partners

01

Validate

Dual-code a sample of narratives for clarity, specificity, need and agency; freeze definitions.

02

Close data gaps

Partner identifiers, all listings including expired ones, original text versions.

03

Pilot

One truthful clarity prompt versus control, with a practical threshold and power agreed in advance.

Thank you

Mike Nguyen and Sophia Nguyen, Cultural Blend

Code and evidence: github.com/tuannm3812/unsw-ma-hackathon-2026



The two-country screen in full

Explanatory models: within-region family-framing slopes in log(1 + days) per extra mention per 100 words

Region	Countries	Loans	Slope	Clustered 95%, normal	t(1) interval	Heuristic p
Middle East	2	14,946	−0.0729	[−0.1003, −0.0456]	[−0.2502, 0.1043]	0.1203
Central America	2	59,391	−0.0742	[−0.0879, −0.0605]	[−0.1629, 0.0145]	0.0597

Why two countries are not enough

A one-country group cannot estimate between-country uncertainty, and two countries estimate it poorly. The t(1) reference keeps the same clustered standard error and widens the interval.

What the screen is

A heuristic sensitivity screen, not a calibrated correction. Passing it would not certify a result, so these patterns stay descriptive.

As-of training audit

Forecast benchmark: training records must have completed before each cutoff

Evaluation year	Pre-cutoff postings completing later	Eligible as-of training	Evaluation rows
2021	7,528	717,672	163,166
2022	3,918	884,448	143,354
2023	3,546	1,028,174	143,233
2024	5,691	1,169,262	141,725
2025	4,261	1,312,417	137,162

Note: The 2024 replay excludes 5,691 training records that a posting-only split would keep. Evaluation windows still select eventual completions.

Chronological holdout metrics

Explanatory models: chronological holdout, completed loans only

5.2031 days

duration MAE, HistGradientBoosting

0.5420

duration R^2

0.8997

24-hour ROC AUC

0.8301

24-hour average precision

0.1156

24-hour Brier score

Scope: 1,174,953 training and 278,887 holdout loans (2024–2025). Retrospective predictions on completed loans, not deployment estimates.

Exploratory story themes

Exploratory NMF themes from a 20,000-loan sample; descriptive, not full-data evidence

1.5 days

mean funding time, fastest theme (sanitation)

13.5 days

mean funding time, slowest theme (group solar and farm plots)

What the themes show

8 themes, including sanitation, clean water, livestock, smallholder farming and retail. They track loan purpose as much as writing style.

How to read them

Means are unadjusted and come from a sample, so the gap is context, not a tested effect of story content.